

# ECE 3210 hw03

1. Compute the following convolutions.

(a)  $y_1(t) = (u(t-1) - u(t-3)) * 2\text{rect}\left(\frac{t}{4}\right)$

(b)  $y_2(t) = (u(t-1) - u(t-3)) * \cos(\pi t)$

2. Compute the following convolutions.

(a)  $y_1(t) = \sin(\pi t)u(t+1) * u(t-2)$

(b)  $y_2(t) = 2(u(t+1) - u(t-1)) * e^{-t}(u(t-1) - u(t-4))$

3. Compute the following convolutions.

(a)  $y_1(t) = \sin(t)(u(t) - u(t - 2\pi)) * u(t)$

(b)  $y_2(t) = u(t) * t(u(t + 1) - u(t - 2))$

4. Compute the following convolutions.

(a)  $y_1(t) = e^{-t}u(t) * (u(t-1) - u(t-3))$

(b)  $y_2(t) = (u(t-1) - u(t-3)) * \cos(\pi t)$

5. Compute the following convolutions.

(a)  $y(t) = 2u(t) * t(u(t) - u(t - 1))$

(b)  $y(t) = \text{rect}\left(\frac{t}{4}\right) * \sin(\pi t) \text{rect}\left(\frac{t}{2}\right)$

6. Compute the following convolution.

(a)  $y_1(t) = e^{-2t}u(t) * e^{-t}u(t)$

7. Determine whether each system is stable, marginally stable, or unstable.

(a)

$$(D^4 + 6D^3 + 9D^2)y(t) = (D - 1)f(t)$$

(b)

$$h(t) = e^{-\frac{t}{2}}u(t)$$

(c)

$$(D^3 + 10D^2 + 26D)y(t) = D^2f(t)$$

8. Determine whether each system is stable, marginally stable, or unstable.

(a)

$$(D^3 + 25D)y(t) = (D^2 + 5)f(t)$$

(b)

$$h(t) = e^{\frac{t}{2}}(u(t) - u(t - 10))$$

(c)

$$(D^6 + 10D^5 + 49D^4 + 110D^3 + 94D^2 - 120D - 144)y(t) = D^2f(t)$$

## Answer Key

$$1(a) \ y_1(t) = \begin{cases} 0, & t < -1, \\ 2(t+1), & -1 \leq t < 1, \\ 4, & 1 \leq t < 3, \\ -2t+10, & 3 \leq t \leq 5, \\ 0, & t > 5. \end{cases}$$

$$1(b) \ y_2(t) = 0$$

$$2(a) \ y_1(t) = \begin{cases} 0, & t < 1, \\ -\frac{\cos(\pi t) + 1}{\pi}, & t \geq 1. \end{cases}$$

$$2(b) \ y_2(t) = \begin{cases} 0, & t < 0, \\ 2e^{-1}(1 - e^{-t}), & 0 \leq t < 2, \\ 2e^{-t}(e - e^{-1}), & 2 \leq t < 3, \\ 2(e^{1-t} - e^{-4}), & 3 \leq t < 5, \\ 0, & t \geq 5. \end{cases}$$

$$3(a) \ y_1(t) = \begin{cases} 0, & t < 0, \\ 1 - \cos(t), & 0 \leq t \leq 2\pi, \\ 0, & t > 2\pi. \end{cases}$$

$$3(b) \ y_2(t) = \begin{cases} 0, & t < -1, \\ \frac{t^2}{2} - \frac{1}{2}, & -1 \leq t < 2, \\ \frac{3}{2}, & t \geq 2. \end{cases}$$

$$4(a) \ y_1(t) = \begin{cases} 0, & t < 1, \\ 1 - e^{-(t-1)}, & 1 \leq t < 3, \\ e^{-t}(e^3 - e^1), & t \geq 3. \end{cases}$$

$$4(b) \ y_2(t) = 0$$

$$5(a) \ y(t) = \begin{cases} 0, & t < 0, \\ t^2, & 0 \leq t < 1, \\ 1, & t \geq 1. \end{cases}$$

$$5(b) \quad y(t) = \begin{cases} 0, & t < -3, \\ -\frac{\cos(\pi t) + 1}{\pi}, & -3 \leq t \leq -1, \\ 0, & -1 < t < 1, \\ \frac{1 + \cos(\pi t)}{\pi}, & 1 \leq t \leq 3, \\ 0, & t > 3. \end{cases}$$

$$6(a) \quad y_1(t) = \begin{cases} 0, & t < 0, \\ e^{-t} - e^{-2t}, & t \geq 0. \end{cases}$$

7(a) unstable

7(b) stable

7(c) marginally

8(a) marginally

8(b) stable

8(c) unstable